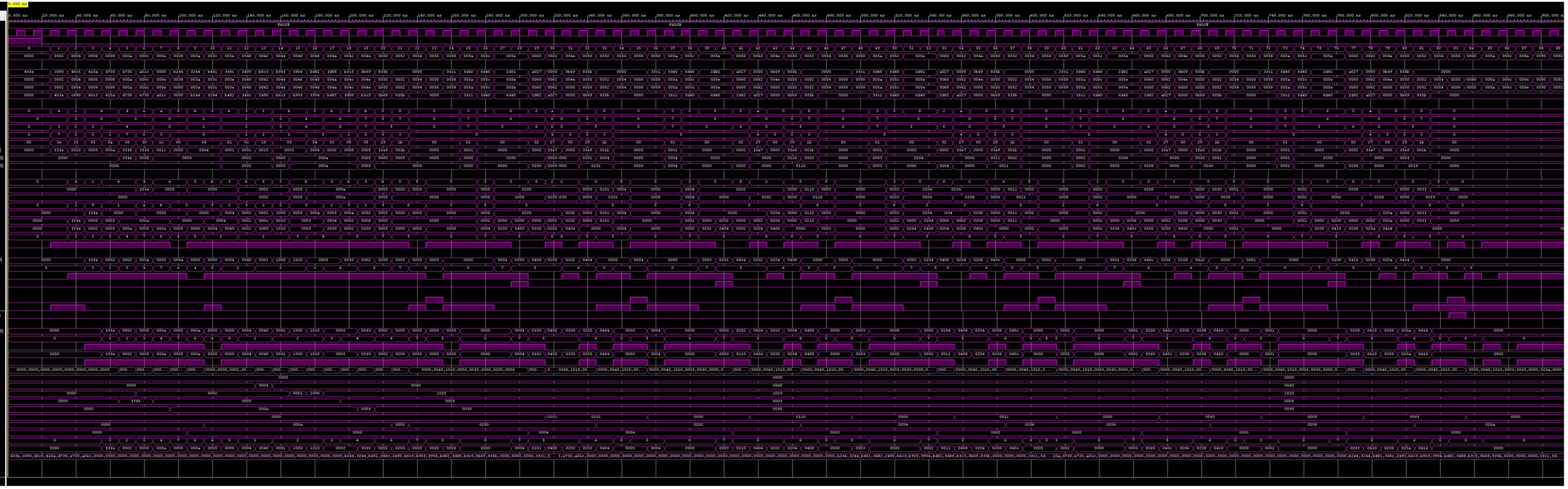
testbench_behav.wcfg

Name

- sim_done
- clk
- reset
- cycle_count
- PC[15:0]
- Instruction Fetch
  - instruction[15:0]
  - IF_PC[15:0]
  - IFID_PC[15:0]
  - IFID_instruction[15:0]
- Instruction Decode
  - ID_opcode[3:0]
  - ID_rs[2:0]
  - ID_rd[2:0]
  - ID_immediate[5:0]
  - ID_sign_extended[15:0]
  - ID_reg_read_data1[15:0]
  - ID_reg_read_data2[15:0]
- Execute
  - IDEX_opcode[3:0]
  - IDEX_read_data1[15:0]
  - IDEX_read_data2[15:0]
  - alu_control[3:0]
  - operand_a[15:0]
  - operand_b[15:0]
  - EX_alu_result[15:0]
  - EX_write_reg[2:0]
    - IDEX_RegWrite
- Memory Stage
  - EXMEM_alu_result[15:0]
  - EXMEM_write_reg[2:0]
    - EXMEM_RegWrite
    - EXMEM_MemRead
    - Branch Decision
      - EXMEM_Branch
      - EXMEM_zero_flag
    - MEM_branch_taken
- Write Back
  - MEMWB_alu_result[15:0]
  - MEMWB_write_reg[2:0]
    - MEMWB_RegWrite
  - WB_write_data[15:0]
    - write_enable
- registers[0:7][15:0]
  - [0][15:0]
  - [1][15:0]
  - [2][15:0]
  - [3][15:0]
  - [4][15:0]
  - [5][15:0]
  - [6][15:0]
  - [7][15:0]
- write_addr[2:0]
- write_data[15:0]
- mem[0:511][15:0]